

Polynomial Regression for Power Plant and Thermal Reservoir Data

Ridge-regularized polynomial models selected by 5-fold cross-validation

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1. Problem Overview

This assignment asks for polynomial regression models for two separately assigned datasets. In both problems the target y is a continuous quantity, and the goal is to predict y accurately on a hidden test set of 1000 samples. The assignment states that the relationships can be represented by polynomials, so polynomial expansion is the natural model class.

Phase 1 (var1): Power Plant Steam Turbine Optimization. Six operational settings determine the *Net Power Score*. A polynomial of degree up to 10 is permitted.

Phase 2 (var2): Subterranean Thermal Reservoir Mapping. Three spatial offsets determine the *Thermal Anomaly Score*, which must be predicted at unobserved locations from scattered 3D observations. A polynomial of degree up to 20 is permitted.

var1	Input description	var2	Input description
x_1	High-pressure steam valve adjustment	x_1	East-west coordinate offset
x_2	Condenser coolant flow rate adjustment	x_2	North-south coordinate offset
x_3	Re-injection pump hydraulic pressure	x_3	Vertical depth offset from basecamp
x_4	Turbine blade pitch angle		
x_5	Non-condensable gas exhaust valve rate		
x_6	Steam inlet pressure adjustment		

Table 1. Input variables of the two problems.

Problem	Inputs	Target y	Max. degree allowed	Degrees searched	Ridge α searched
var1	6	Net Power Score	10	4, 5, 6	1, 3, 10
var2	3	Thermal Anomaly Score	20	10, 11, 12, 13, 14	0.03, 0.1, 0.3

Table 2. Summary of the two problems and the hyperparameter search ranges.

2. Overall Approach

The same pipeline is applied independently to each dataset. The inputs are expanded into polynomial features, a Ridge (L2-regularized) linear model is fitted on those features, and the polynomial degree and the regularization strength α are chosen together by 5-fold cross-validation. The pair with the smallest average validation MSE is kept, the model is retrained on all labelled training data, and the result is used to predict the test targets. The hidden test labels play no role in model selection.

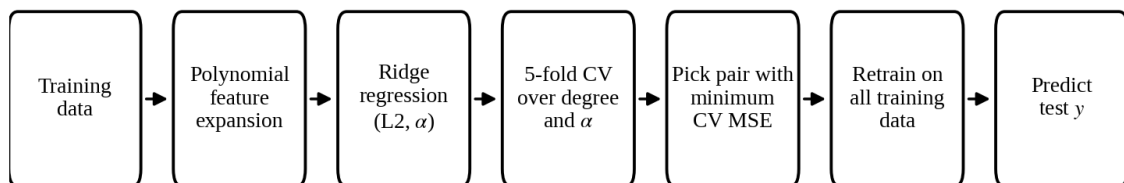


Figure 1. Modelling pipeline used for both var1 and var2.

3. Methodology

3.1 Polynomial regression

A polynomial model gives a nonlinear response in the original variables while remaining *linear in its coefficients*. For two inputs and degree 2 the model is

$$y = w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_1x_2 + w_5x_2^2$$

A polynomial of degree d contains every term whose powers sum to at most d . For example $x_1^3x_2^2$ has total degree 5 and is therefore included in a degree-5 model. Because the model is linear in the coefficients, an ordinary linear solver can be applied to the polynomial feature matrix. The expansion grows quickly with degree and number of inputs: the selected var1 model (6 inputs, degree 5) has 461 polynomial terms and the selected var2 model (3 inputs, degree 12) has 454, both excluding the intercept.

3.2 Loss function

Fitting is based on the squared error. Averaged over N samples this is the Mean Squared Error,

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

MSE is also the primary metric for choosing degree and α ; lower means closer predictions on average.

3.3 Ridge (L2) regularization

High-degree polynomials have many coefficients and can fit noise as well as signal. Ridge regression limits this by adding an L2 penalty on the coefficient vector:

$$J(w) = \|y - Xw\|_2^2 + \alpha \|w\|_2^2$$

The first term measures the fit to the data and the second penalizes large coefficients. The parameter α sets the strength of the penalty: a larger α shrinks the coefficients more strongly, while a smaller α gives the model more freedom to fit. L2 shrinks coefficients toward zero without forcing them exactly to zero, which suits this task because the aim is to stabilize the polynomial model rather than to select features.

3.4 Cross-validation for model selection

Model selection uses `KFold(n_splits=5, shuffle=True, random_state=42)`. The training set is shuffled and split into five folds; in each round the model is trained on four folds and its MSE is measured on the remaining fold. The five validation MSEs are averaged, and the configuration with the smallest average is selected. Fixing `random_state=42` makes the folds identical on every run, so the experiment is repeatable.

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Round 1	validate	train	train	train	train
Round 2	train	validate	train	train	train
Round 3	train	train	validate	train	train
Round 4	train	train	train	validate	train
Round 5	train	train	train	train	validate

Figure 2. The five CV rounds: each fold is used once for validation (shaded) and four times for training.

Degree and α are searched *jointly*: every (degree, α) pair is scored on all five folds, and the best pair is kept.

3.5 Final retraining

Cross-validation only chooses hyperparameters. Within a fold each model sees 80% of the training data, so the selected model is refit on the *complete* training set and this final model produces the test predictions.

4. Model Selection and Results

Table 3 reports the selected configuration for each problem together with its cross-validation and training statistics. The two problems were tuned independently.

Problem	Degree	Ridge α	CV MSE	CV R^2	Train MSE	Train R^2	Terms
var1	5	3	0.423664	0.955142	0.182297	0.980991	461
var2	12	0.1	0.248077	0.995260	0.161017	0.996939	454

Table 3. Selected hyperparameters and performance. CV values are 5-fold averages; “Terms” is the number of polynomial features excluding the intercept.

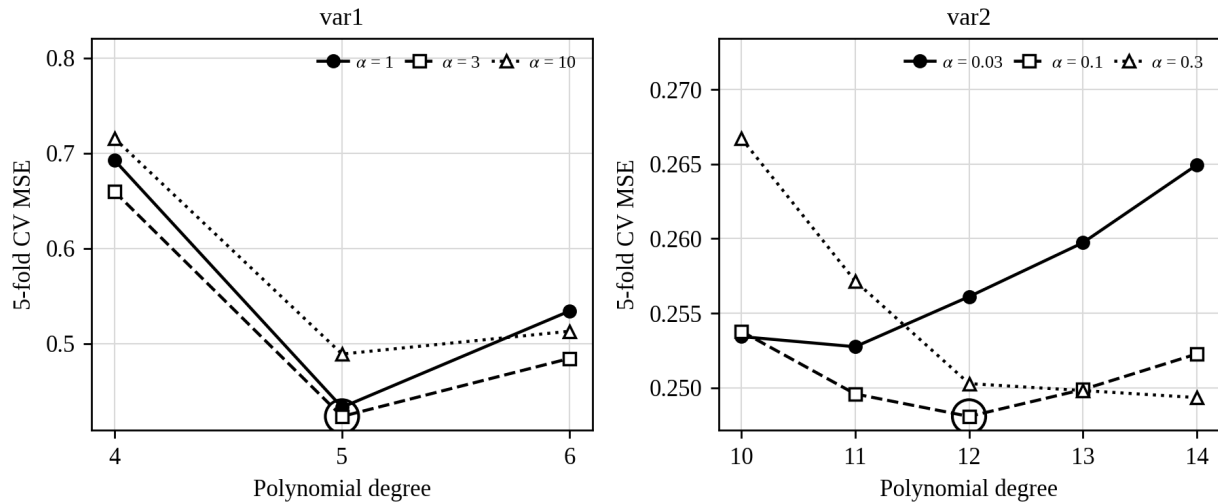


Figure 3. Average 5-fold CV MSE against polynomial degree for each tested α . The circled point marks the selected configuration in each problem.

4.1 var1: degree 5, $\alpha = 3$

Degree 4 gives a clearly higher validation error (0.660 at its best α), so it does not capture enough of the nonlinear structure. Degree 6 does not improve on degree 5 for any tested α (best 0.485). The minimum, a CV MSE of 0.423664 at $\alpha = 3$, therefore lies at degree 5. The mean CV R^2 of about 0.955 compares with a training R^2 of about 0.981, as expected for a model fitted directly on the training observations.

4.2 var2: degree 12, $\alpha = 0.1$

Validation error falls as the degree rises from 10 to 12 and then no longer improves for the same α . The lowest value, 0.248077, occurs at degree 12 with $\alpha = 0.1$. The differences among the leading candidates are small (for example 0.2493 at degree 14 with $\alpha = 0.3$), which indicates a flat optimum; the configuration with the smallest average CV MSE was used. The mean CV R^2 of about 0.995 means the model explains nearly all of the variation in the validation folds.

5. Discussion

var1				var2			
Degree	$\alpha = 1$	$\alpha = 3$	$\alpha = 10$	Degree	$\alpha = 0.03$	$\alpha = 0.1$	$\alpha = 0.3$
4	0.692961	0.659964	0.715787	10	0.253433	0.253786	0.266737
5	0.433218	0.423664	0.489167	11	0.252751	0.249584	0.257157
6	0.534393	0.484593	0.513144	12	0.256111	0.248077	0.250279
				13	0.259729	0.249881	0.249813
				14	0.264946	0.252267	0.249341

Table 4. Complete 5-fold CV MSE grids (minimum in bold).

Same method, different hyperparameters. Both problems use the pipeline polynomial features → Ridge → 5-fold CV, but the optimal settings differ because the datasets differ. var1 has six inputs, so even degree 5 already produces 461 terms with many interaction terms, and a stronger penalty ($\alpha = 3$) works best. var2 has three inputs, which allows a degree-12 polynomial with a comparable number of terms (454) and a much weaker penalty ($\alpha = 0.1$). Nothing requires the two problems to share the same degree or shrinkage.

Underfitting and overfitting. A degree that is too low cannot represent the true curvature and interactions; a degree that is too high fits noise in the training data. Training error alone cannot tell these apart because it keeps falling as complexity grows. Cross-validation measures error on held-out data, and the L2 penalty further discourages excessively large coefficients. For this reason the final models use neither the highest degree allowed (10 and 20) nor the lowest.

Reading MSE and R^2 . MSE is the average squared prediction error (lower is better) and was the criterion for selecting the (degree, α) pair. The coefficient of determination,

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

compares the model with a baseline that always predicts the mean (higher is better). It is reported as a supporting statistic. Because MSE and R^2 are not optimized identically across folds, a different configuration can occasionally show a slightly higher mean R^2 ; the selection rule used here is the minimum average CV MSE.

Training versus validation. Training MSE (0.182 for var1, 0.161 for var2) is below CV MSE (0.424 and 0.248) because the final model is fitted on the training points themselves. The CV figures are the better estimate of performance on unseen data.

6. Predictions and Conclusion

With the selected configurations, one final model per problem was fitted to all training data and used to predict the 1000 test samples. The outputs are BT2024266_pred_var1.csv and BT2024266_pred_var2.csv, each with the header y followed by 1000 predictions.

In summary, Ridge-regularized polynomial regression with joint, cross-validated selection of degree and α gave a degree-5, $\alpha = 3$ model for var1 (CV MSE 0.4237, CV R^2 0.955) and a degree-12, $\alpha = 0.1$ model for var2 (CV MSE 0.2481, CV R^2 0.995). Hyperparameters were chosen without using the test labels, and the final models use all available training data.